

## Day 6 Assignment – Triage Pseudocode

As a designer of these AI systems your role is to pull it together by describing through ANY method of your choice (code-based, instruction-based, flowcharts, paragraphs etc) how a digital system can process patient data and categorize them into a risk level.

### Data Input and Cleaning

Before the system calculates any risk levels, it has to clean and validate the incoming patient data. In an ED setting, data is gathered through manual entry into a tablet by the triage nurse or through direct Bluetooth syncing from connected patient monitors like pulse oximeters and blood pressure cuffs.

- Demographics: Age, Gender (0, 1, 2)
- Vital Signs: Respiratory Rate (RR), Pulse (Heart Rate), Systolic Blood Pressure (SBP), Fraction of Inspired Oxygen (FiO2)
- Neurological Status: Glasgow Coma Scale (GCS)
- Other Vitals: Oxygen Saturation (SpO2), Capillary Blood Glucose (CBG), Capillary Refill Time (CRT)

### Checking Physiological Boundaries

To prevent typos or machine reading errors from throwing off the clinical logic, all incoming vitals are checked against extreme human limits (e.g., a heart rate of 1000 bpm or an SpO2 over 100%). If a value falls outside these realistic boundaries, the system strips the bad data and marks it as NaN.

### Handling Missing Data

In an emergency room, the system cannot afford to stall or drop a patient profile just because a vital sign is missing. It can handle missing data using two backup steps:

1. **Hospital EHR Search** – the system first searches the hospital’s Electronic Health Records to see if the specific patient has a historical baseline from previous visits.
2. **Population Median Fallback** – if there’s no history, the system fills the gap using a safe population median (e.g., setting a missing respiratory rate to 18 breaths/min). Medians are used instead of averages so that data isn’t skewed by a few critically ill patients (outliers).

**Safety Alert** – whenever missing data is filled in via Tier 2, a warning “Imputed Data Used – Verify Manually” will appear on the nurse’s screen so they know to physically re-measure that vital as soon as possible.

### Clinical Assessment Logic

**Visual Trauma Emergency Override** – before running any numerical scoring, the system checks a single priority box at the top of the interface: Is\_Visual\_Trauma\_Emergency. If this is checked (for visible, life-threatening emergencies like active major bleeding, gunshot/stab wounds or a blocked airway), the system instantly bypasses all calculations and assigns the patient straight to Level 1 – Resuscitation.

**Dynamic Baseline Modifiers** – to prevent false alarms, the system looks at a patient’s medical history and current medications before scoring their vitals. E.g., if a person has chronic lung disease (COPD), their body naturally works with lower oxygen levels (88% to 92%). The system automatically recognises

this and switches to a personalised scoring chart for that patient so it doesn't trigger a false emergency alert.

**Vital Interactions** – vitals are not looked at in isolation because interacting symptoms can indicate a crashing patient. The system checks for critical multi-variable combinations. E.g., Cardiorespiratory Panic – if the heart and lungs are fighting together. Is Pulse > 100 bpm AND RR > 20 breaths/min? If yes, the patient is in potential systemic shock. They are labelled high risk in a triage system.

### Points Scoring

If no immediate visual bypass is triggered, the system starts the patient's Total\_Score (Total Clinical Urgency Score) at 0 and adds points based on how far their vitals stray from normal ranges. For example:

- Respiratory Distress Scaling (RR in breaths/min)
  - If RR is between 12 and 20 → add 0 points (normal baseline)
  - If RR is between 21 and 24 → add 1 point (mild tachypnea)
  - If RR > 24 (severe tachypnea) OR RR < 12 (Bradypnea) → add 3 points
  - If RR < 8 → add 5 points (imminent respiratory arrest)
- Oxygen Saturation (SpO2 in %)
  - If SpO2 < 90% → add 5 points (severe hypoxia emergency)
    - However, in persons with compromised lungs, etc surgically removed, the baseline would drop. AI should note.
  - If SpO2 is between 91% and 94% → add 2 points (moderate hypoxia)
- Neurological Status (GCS Scale)
  - If GCS is between 13 and 15 → add 0 points (normal/mild)
  - If GCS is between 9 and 12 → add 3 points (moderate neurological/consciousness impairment)
  - If GCS < 8 → add 5 points (severe brain injury/unconscious. Less than 8 might have to intubate)

### Triage Risk Categorisation

Once the points are added up, the system sorts the patient into one of four priority levels that dictate how quickly they must be seen by clinical staff.

| Total Score   | Risk Category           | Clinical Meaning                                                                       |
|---------------|-------------------------|----------------------------------------------------------------------------------------|
| >= 7 points   | Level 1 – Resuscitation | Immediate threat to life. Continuous monitoring required.                              |
| 4 – 6 points  | Level 2 – Emergent      | High-acuity patient. Potential for rapid decline. Assessment within 15 minutes.        |
| 2 to 3 points | Level 3 – Urgent        | Stable vitals but clear systemic illness. Assessment within 30 – 60 minutes.           |
| 0 to 1 point  | Level 4 – Non-Urgent    | Normal physiology. Safe to remain in the waiting pool with routine re-triage tracking. |

## Fail-Safes and Waiting Room Management

**Power Outage Shield** – to handle infrastructure or electricity failures, which are common in overcrowded Caribbean ED environments, the AI model does not rely solely on the cloud. It uses a Local-First architecture with an encrypted SQLite database on the tablet itself. Every time data is entered, it saves directly to the tablet's hardware storage before trying to sync over Wi-Fi. If the power or Wi-Fi goes down, the tablet runs seamlessly offline on battery power, and automatically syncs back up to the hospital servers once the connection returns.

**Digital Aging for Waiting Patients** – because waiting rooms can become overcrowded and understaffed, patients left sitting for long periods run the risk of declining unnoticed. The model uses a time-based decay function. For every 15 minutes a patient remains in the waiting room, the system automatically adds +0.5 points to their priority score. This pushes them up the queue to force a mandatory physical re-evaluation by a nurse.

**Computer Vision Monitoring** – to support busy nurses who cannot constantly watch the waiting room, the system links with privacy-compliant waiting room cameras running an anomaly detection network. If the camera detects a patient suddenly slumping over, falling from a seat, clutching their chest or other visual cues, it immediately sets off a Critical Escalation Alarm on the nurse's primary console and forces the patient's profile straight to Level 1.

**System Crash Fallback (Human-in-the-Loop)** – to ensure the AI remains a helpful tool and never a single point of failure, the entire calculation script is wrapped in a safety catch (try-except block). If a software error or database crash happens, the system is blocked from freezing or leaving a patient unregistered. Instead, it throws an immediate alert reading "AI Disconnected / Manual Triage Required" and switches the interface over to a traditional, manual scoring layout. This guarantees that nurses always hold absolute veto power and final diagnostic authority.

## AI Model Instruction-Based/Flowchart Style

[INPUT] → Raw Demographics, Vitals, History & Visual Flags

- Age, Gender, RR, Pulse, SBP, SpO2, GCS, CBG, CRT, Medical History (Maybe Patient ID)

# Data can be gathered via direct Bluetooth telemetry sync from patient monitors where possible, or manually input into the tablet interface by the triage nurse.

[GLOBAL TRY-EXCEPT BLOCK] → System-Wide Failure Safeguard

- Try: Execute STAGE 1 through OUTPUT
- Except (Any Error/System Crash): Trigger immediate UI Alert: "AI Disconnected / Manual Triage Required"

- Instantly fall back to standard manual triaging

# This ensures the app never freezes or leaves a patient unregistered during database errors or unhandled software exceptions, leaving absolute veto power to nurses

[STAGE 1] → Intermediate Visual Trauma Screening

- Check: Is\_Visual\_Trauma\_Emergency == True?
  - YES: Instantly route to LEVEL 1 – Resuscitation (Bypass all)
  - NO: Continue to STAGE 2

# This allows the nurse to flag patients arriving with visible, life-threatening emergencies (e.g., active arterial bleeding, gunshot/stab wounds or compromised airways) so they bypass numerical calculations entirely

[STAGE 2] → Range Filtering & Verification

- STEP A: Physiological Boundary Check
  - Evaluate incoming vitals against extreme, impossible limits (e.g., Pulse > 600 bpm, RR > 60 breaths/min, or SpO2 > 100%)
  - If a vital breaks these boundaries → Strip the value and set to NaN (Data Entry Error)
- STEP B: Two-Tier Imputation. For any NaN or missing vital:
  - TIER 1 (Local Lookup): Search the patient's local Electronic Health Record (EHR) for an individual baseline.
  - TIER 2 (Fallback): If no history exists, fill the gap using the population median (e.g. set missing RR = 18 breaths/min)
- STEP C: User Interface Notification
  - If any vital was imputed → Append "Imputed Data Used – Verify Manually" warning on the nurse's screen

# Missing data and typos can completely affect machine learning models and scoring logic. This safely cleans the data using robust population medians rather than skewed means, while alerting the nurse to double check

[STAGE 3] → Baseline Customisation Lookups

- Check history: Does patient have history of severe lung disease or COPD?
  - Yes: Swap out standard SpO2 metrics for Personalised Scoring Limits (SpO2 normal baseline reset to 88% – 92% for that patient).
  - No: Keep default clinical ranges

# Patients with chronic conditions like COPD safely walk around with lower oxygen levels. This step prevents false-positive emergencies.

# Include similar diseases with different baselines, other than COPD.

[STAGE 4] → Clinical Combo Assessment

- Check multi-variable interactions: E.g., Is Pulse > 100 AND RR > 20?
    - Yes: Apply +3 points to the Total Clinical Urgency Score
  - Check severe interaction: E.g., Is Pulse > 120 AND RR > 24?
    - Yes: Apply +5 points to the Total Clinical Urgency Score
- # The interface can also include diagrammatic visualisations like scatter plots to better display this for triage nurses. Looking at vitals together catches patients whose internal organs are shutting down systematically

[STAGE 5] → Point Aggregation

- Initialise Total\_Score = 0. Add points based on the clinical deviations:
    - Respiratory Rate (RR)
      - 12 to 20 breaths/min → +0 points (healthy adult baseline)
      - 21 to 24 breaths/min → +1 point (mild tachypnea)
      - < 12 OR > 24 breaths/min → +3 points (bradypnea or severe tachypnea)
      - < 8 breaths/min → +5 points (imminent respiratory arrest)
    - Oxygen Saturation (SpO<sub>2</sub>)
      - 91% to 94% → +2 points (moderate hypoxia)
      - ≤ 90% → +5 points (severe hypoxia/medical emergency)
    - Neurological Status (GCS)
      - 13 to 15 → + 0 points (mild/intact)
      - 9 to 12 → +3 points (moderate consciousness impairment)
      - ≤ 8 → +5 points (severe injury/unconscious - <8 intubate)
    - Capillary Blood Glucose (CBG)
      - < 54 mg/dL → +4 points (severe hypoglycemia mimicking a stroke/coma)
      - > 250 mg/dL → +3 points (severe hyperglycemia/potential DKA crisis)
    - Capillary Refill Time (CRT)
      - ≥ 3 seconds → +2 points (poor peripheral blood perfusion / circulatory shock indicator)
- # Consider age (Newborns, Adults and Older Adults have different ranges)
- # Any other metric. This block is the central clinical logic matrix, converting single continuous variables into a risk score.

[OUTPUT] → Categorise into Tier

- Evaluate the final Total\_Score to assign a definitive Triage Level
  - Total\_Score ≥ 7 → LEVEL 1 – RESUSCITATION (immediate threat to life)
  - Total\_Score between 4 and 6 → LEVEL 2 – EMERGENT (high acuity; potential to rapidly decline; assess within 15 mins)
  - Total\_Score between 2 and 3 → LEVEL 3 – URGENT (stable physiology but systemically ill; assess within 30-60 minutes)
  - Total\_Score between 0 and 1 → LEVEL 4 – NON-URGENT (normal physiology; safe to wait with routine tracking)
- STEP A: Power Outage Hardware Shield
  - Store patient's finalised score and data directly to the device's internal, encrypted SQLite local hardware storage prior to trying to push it over Wi-Fi
  - If hospital Wi-Fi/power fails, the tablet operates offline using battery power. It automatically performs a background database sync once connection is restored

- STEP B: Waiting Room Digital Aging
  - Track queue timestamps. For every 15 minutes a patient remains waiting, automatically add +0,5 points to their active priority score to move them up the queue
- STEP C: Computer Vision Integration
  - Stream privacy-compliant waiting room camera feeds through a posture-anomaly network
  - If a waiting patient suddenly slumps over, falls from their seat, or clutches their chest (or other visual cues) → automatically sound a "Critical Escalation Alarm" on the nurse console and force an immediate shift to LEVEL 1

# The final output phase ensures the system functions safely in a Caribbean environment by accounting for power cuts, busy waiting lines and patient deterioration over time